

3.1.2 Monotonicity Worksheet

1. Show that the sequence $a_n = \frac{2}{n}$ is strictly decreasing.

I will use the ratio of consecutive terms to show the sequence is strictly decreasing.

$$\frac{a_{n+1}}{a_n} = \frac{\left(\frac{2}{n+1}\right)}{\left(\frac{2}{n}\right)} = \frac{2n}{2n+2} < 1 \text{ for all } n.$$

This implies that $a_{n+1} < a_n$ for all n , therefore the sequence is strictly decreasing.

2. Determine whether the sequence $a_n = \frac{3^n}{n!}$ is eventually increasing or decreasing.

I will use the ratio of consecutive terms to show that the sequence is eventually increasing or decreasing.

$$\frac{a_{n+1}}{a_n} = \frac{\left(\frac{3^{n+1}}{(n+1)!}\right)}{\left(\frac{3^n}{n!}\right)} = \frac{3^{n+1}n!}{3^n(n+1)!} = \frac{3}{n+1} \leq 1 \text{ for } n \geq 2$$

This implies that $a_{n+1} \leq a_n$ for $n \geq 2$, therefore the sequence is eventually decreasing.

3. Show that the sequence $a_n = \frac{n}{n+2}$ is strictly increasing using the derivative/calculus technique.

Let $f(x) = \frac{x}{x+2}$, then $f'(x) = \frac{(x+2) - x}{(x+2)^2} = \frac{2}{(x+2)^2}$. We note that for $x \geq 1$, $f'(x) > 0$ and therefore

$f(x)$ is increasing for $x \geq 1$. Therefore, $a_n = \frac{n}{n+2}$ is increasing for $n \geq 1$.

4. Show how the Monotone Sequence Theorem can be used to show that the sequence $\{a_n\}_{n=2}^{\infty}$, with general term $a_n = \frac{2n+3}{n-1}$, converges.

To use the Monotone Sequence Theorem we must show that the sequence $\{a_n\}_{n=2}^{\infty}$ is decreasing and bounded below or increasing and bounded above.

We will use calculus to show that the given sequence is decreasing.

Let $f(x) = \frac{2x+3}{x-1}$, then $f'(x) = \frac{(x-1)2 - (2x+3)}{(x-1)^2} = \frac{-5}{(x-1)^2}$. We note that for $x \geq 2$, $f'(x) < 0$ and

therefore $f(x)$ is decreasing for $x \geq 2$. Therefore, $a_n = \frac{2n+3}{n-1}$ is decreasing for $n \geq 2$.

We can also see that for $n \geq 2$, $a_n = \frac{2n+3}{n-1} > 0$, therefore the sequence $\{a_n\}_{n=2}^{\infty}$ is bounded below by 0.

Since $\{a_n\}_{n=2}^{\infty}$ is decreasing and bounded below, then by the Monotone Sequence Theorem it must converge.